MATHEMATICAL ANALYSIS 1
HOMEWORK 6

Note. This week we covered a lot of material, so this homework contains more problems than usual. For the moment it is enough to solve Problems 1–7. By the end of the semester you should also solve the Bonus problems and the Problems for later at the end of this sheet.

(1) **Indeterminate forms of algebraic type.** Compute the following limits. Explain your answers.

(a) $\lim_{x\to 0} \dfrac{x^4 - 2x^3 + 5x}{x^5 - x}$

(b) $\lim_{x\to -1} \dfrac{x+1}{\sqrt{6x^2+3}+3x}$

(c) $\lim_{x\to +\infty} \left(\sqrt{x+1} - \sqrt{x}\right)$

(d) $\lim_{x\to -\infty} \dfrac{3^x - 3^{-x}}{3^x + 3^{-x}}$

(e) $\lim_{x\to +\infty} \dfrac{x+3}{x^3 - 2x + 5}$

(f) $\lim_{x\to 0} \dfrac{\sqrt{1+\tan x} - \sqrt{1-\tan x}}{\sin x}$

(g) Determine the behavior of $f(x) = \frac{x^3 - x^2 + 3}{x^2 + 3x + 2}$ as $x \to +\infty$ and as $x \to -\infty$. (In Homework 5, Problem 9, you studied this function near $x = -2$ and $x = -1$.)

(2) **The Substitution Theorem.** Whenever you use the Substitution Theorem (Theorem 6.12 of the lecture notes), say which functions play the roles of f and g, which of the three cases of the theorem you use, and check its hypotheses.

(a) Compute $\lim_{x\to+\infty} x \sin\frac{1}{x}$ and $\lim_{x\to 0} x \sin\frac{1}{x}$. Why do the two limits require different tools?

(b) Deduce the value of $\lim_{x\to+\infty} \left(5 + 2x \sin\dfrac{1}{x}\right)$.

(c) Compute $\lim_{x\to 1} \dfrac{\cos\left(\frac{\pi}{2}x\right)}{1-x}$. *Hint: take $y = 1 - x$.*

(d) Compute the following limits (one-sided where indicated):

$$\lim_{x\to 0} e^{-1/x^2}, \qquad \lim_{x\to 0^+} e^{1/x}, \qquad \lim_{x\to 0^-} e^{1/x},$$

$$\lim_{x\to 0^+} \ln(\sin x), \qquad \lim_{x\to 0^+} \arctan\frac{1}{x}, \qquad \lim_{x\to 0^-} \arctan\frac{1}{x}.$$

(3) **Limits of sequences.** Determine how the following sequences $\{a_n\}_{n\in\mathbb{N}_+}$ behave as $n \to \infty$ (converge, diverge to $\pm\infty$, or are indeterminate). Explain your answers.

(a) $a_n = n - \sqrt{n}$

(b) $a_n = \frac{(2n)!}{n!}$

(c) $a_n = \frac{(2n)!}{(n!)^2}$

(d) $a_n = \binom{n}{3}\frac{6}{n^3}$

(e) $a_n = 2^n \sin\left(2^{-n}\pi\right)$

(f) $a_n = n\cos\left(\frac{n+1}{n} \cdot \frac{\pi}{2}\right)$

(4) **Fundamental limits.** Compute the following limits. Explain your answers.

(a) $\lim_{x\to 0} \dfrac{x - \tan x}{x^2}$ *(hint: Squeeze Theorem, and the inequality $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$ from Example 5.13 of the lecture notes)*

(b) $\lim_{x\to e} \dfrac{\ln x - 1}{x - e}$ *(hint: take $y = x - e$)*

(c) $\lim_{x\to 0^+} \dfrac{2^{2x} - 1}{2x}$

(d) $\lim_{x\to 1} \dfrac{\ln x}{e^x - e}$ *(hint: take $y = x - 1$)*

(e) $\lim_{x\to 0} \dfrac{e^x - e^{-x}}{\sin x}$

(f) Determine the domain of $f(x) = \sqrt[3]{x}\, e^{-x^2}$ and the behavior of f at the end-points of its domain. *(Hint: you may use the fact that* $\lim_{x\to+\infty}\left(\frac{1}{3}\ln x - x^2\right) = -\infty$*.)*

(5) **Two applications.**

(a) *Radioactive decay.* A radioactive sample contains $N(t) = N_0\, 2^{-t/T}$ atoms at time $t \geq 0$, where $N_0 > 0$ is the initial number of atoms and $T > 0$ is the half-life.

(i) Compute the initial rate of decay $\lim\limits_{h\to 0^+} \dfrac{N(h) - N_0}{h}$.

(ii) Show that $\lim\limits_{n\to\infty} n\left(2^{1/n} - 1\right) = \ln 2$.

(b) *Charging a capacitor.* In an RC circuit (resistance $R > 0$, capacitance $C > 0$) connected at time $t = 0$ to a battery of voltage $V_0 > 0$, the voltage across the capacitor is $V(t) = V_0\left(1 - e^{-t/(RC)}\right)$ for $t \geq 0$.

(i) Compute the final voltage $\lim_{t\to+\infty} V(t)$.

(ii) Compute the initial rate of charging $\lim\limits_{t\to 0^+} \dfrac{V(t)}{t}$.

(iii) The number $\tau = RC$ is called the *time constant* of the circuit. What fraction of the final voltage has been reached at time $t = \tau$?

(6) **Understanding Landau symbols.** Justify each answer using the definitions of the Landau symbols.

(a) Consider the function $4 + \frac{\sin x}{x}$ as $x \to \pm\infty$.

(i) Show that $\frac{\sin x}{x} = o(x)$ as $x \to \pm\infty$, and deduce that $4 + \frac{\sin x}{x} = 4 + o(x)$ as $x \to \pm\infty$.

(ii) Show that also $\frac{\sin x}{x} = o(1)$ and $\frac{\sin x}{x} = \mathcal{O}\left(\frac{1}{x}\right)$ as $x \to \pm\infty$.

(iii) Which of the three statements $4+\frac{\sin x}{x} = 4+o(x)$, $4+\frac{\sin x}{x} = 4+o(1)$, $4+\frac{\sin x}{x} = 4+\mathcal{O}\left(\frac{1}{x}\right)$ gives the most information? Give an example of a function f with $f(x) = 4 + o(x)$ as $x \to +\infty$ which does not tend to 4.

(b) True or false, as $x \to +\infty$?

$$x^2 + 3x = x^2 + o(x^2), \quad x^2 + 3x = \mathcal{O}(x), \quad \sqrt{x} = o(x), \quad x + \sin x \sim x, \quad x + \sin x = x + \mathcal{O}(1), \quad x^2 = o(x^3).$$

(c) True or false, as $x \to 0$? (Compare with (b): near 0 the larger powers are the negligible ones.)

$$x^2 + x^3 = o(x), \quad x^3 = o(x^2), \quad x = o(x^2), \quad \sin x = x + o(x),$$
$$1 - \cos x = o(x), \quad e^x = 1 + x + o(x), \quad \ln(1 + x) = x + o(x).$$

(d) *Warnings.*

(i) Show that $x^2 + x \sim x^2$ as $x \to +\infty$, but that $(x^2 + x) - x^2$ does not tend to 0. So $f \sim g$ does not mean that $f - g \to 0$.

(ii) As $x \to 0$ we have $x^2 = o(x)$ and $2x^2 = o(x)$, but $x^2 \neq 2x^2$. Explain why "$o(g) - o(g) = 0$" is wrong.

(e) *Relativistic kinetic energy.* A body of mass m moving with speed v has kinetic energy $(\gamma - 1)mc^2$, where c is the speed of light, $\beta = \frac{v}{c}$ and $\gamma = \left(1 - \beta^2\right)^{-1/2}$. Using $\lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$ and the Substitution Theorem, show that $\gamma = 1 + \frac{\beta^2}{2} + o(\beta^2)$ as $\beta \to 0$. Conclude that $(\gamma - 1)mc^2 = \frac{1}{2}mv^2 + o(v^2)$ as $v \to 0$: at low speeds we recover the classical kinetic energy.

(7) **Computing with little o.** Use Proposition 8.1 and Corollary 8.2 of the lecture notes and the comparison of powers, and say at each step which rule you use. Recall that such identities are read from left to right.

(a) Show that, as $x \to 0$,

$$\left(x + 3x^2 + o(x^2)\right)\left(2 - x + o(x)\right) = 2x + 5x^2 + o(x^2).$$

Can you also determine the coefficient of x^3 in this product? Why or why not?

(b) *Thermal expansion.* A metal cube has side length $L_0 > 0$ at a reference temperature θ_0. When the temperature changes to $\theta_0 + T$, its side length is $L(T)$. Here T may be positive (heating) or negative (cooling). Experiments show that

$$L(T) = L_0\left(1 + \alpha T + o(T)\right) \qquad \text{as } T \to 0,$$

where $\alpha > 0$ is the coefficient of linear expansion of the metal (for steel, $\alpha \approx 1.2 \cdot 10^{-5}$ K^{-1}). Show that the area of a face is $A(T) = L_0^2\,(1 + 2\alpha T + o(T))$ and the volume is $V(T) = L_0^3\,(1 + 3\alpha T + o(T))$ as $T \to 0$. In words: for small temperature changes, the relative change of the area is about twice, and that of the volume about three times, the relative change of the length.

(c) *Electrical power.* As $t \to 0$, the voltage and the current in a circuit are $V(t) = V_0\,(1 + 2t + o(t))$ and $I(t) = I_0\,(1 - t + o(t))$. Show that the power $P = VI$ satisfies $P(t) = V_0 I_0\,(1 + t + o(t))$ as $t \to 0$. In words: for small t, the relative change of the power is approximately the sum of the relative changes of the voltage and of the current.

(d) *Precision is limited by the worst term.* Show that, as $x \to 0$, $(2 + o(1))\,\big(x - x^3 + o(x^3)\big) = 2x + o(x)$. Explain why we cannot conclude that the product equals $2x - 2x^3 + o(x^3)$.

(e) *Subtracting does not cancel the o's.* Show that as $x \to 0$ $\big(x + 2x^2 + o(x^2)\big) - \big(x + 2x^2 + o(x^3)\big) = o(x^2)$. Why is the answer not 0?

(f) *At infinity the roles are reversed.* Show that, as $x \to +\infty$,

$$\big(x^2 + x + o(x)\big)\left(1 + \frac{1}{x} + o\left(\frac{1}{x}\right)\right) = x^2 + 2x + o(x).$$

Hint: as $x \to +\infty$, every constant is $o(x)$, and $o(1) = o(x)$.

Bonus problems

(B1) **The Ratio Test for sequences.** In this problem we prove Theorem 6.16 of the lecture notes (you are guided in the steps below). Let $\{a_n\}_{n\in\mathbb{N}}$ be a sequence with $a_n > 0$ for all large n, and suppose that the limit $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = q$ exists (finite or $+\infty$). You may use that $\lim_{k\to\infty} r^k = 0$ if $0 < r < 1$, and $\lim_{k\to\infty} r^k = +\infty$ if $r > 1$.

(a) State Theorem 6.16.

(b) Suppose that $q < 1$. Explain why $q \geq 0$ (*hint: Theorem 6.15, item 4*), and choose a number r with $q < r < 1$.

(c) Use the definition of the limit with $\varepsilon = r - q$ to show that there exists $N \in \mathbb{N}$ such that $a_n > 0$ and $a_{n+1} < r\, a_n$ for all $n > N$.

(d) Let $M = N + 1$. Prove by induction on k that $0 < a_{M+k} \leq r^k a_M$ for all $k \in \mathbb{N}$.

(e) Conclude that $\lim_{n\to\infty} a_n = 0$ (*hint: Theorem 6.15, item 5(a)*).

(f) Now suppose that $q > 1$ (including $q = +\infty$). Choose r with $1 < r < q$ (for instance $r = 2$ if $q = +\infty$), show as in (c) and (d) that $a_{M+k} \geq r^k a_M$ for all $k \in \mathbb{N}$, and conclude that $\lim_{n\to\infty} a_n = +\infty$ (*hint: Theorem 6.15, item 5(b)*).

(g) Give three sequences with $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 1$ whose limits are 0, 1 and $+\infty$, respectively. Why does this show that no conclusion is possible when $q = 1$?

(h) Use the theorem to compute $\displaystyle\lim_{n\to\infty} \frac{n^{10}}{1.1^n}$ and $\displaystyle\lim_{n\to\infty} \frac{100^n}{n!}$. (An algorithm whose running time grows like 1.1^n is therefore eventually slower than one whose running time grows like n^{10}, even though for $n = 100$ we have $1.1^{100} \approx 1.4 \cdot 10^4$ while $100^{10} = 10^{20}$.)

(B2) **Proofs with Landau symbols.** Throughout this problem $f, g, h : \mathbb{R} \to \mathbb{R}$, $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$, every function that appears in a denominator is nonzero on a deleted neighborhood of x_0, and all statements are as $x \to x_0$.

(a) (i) Prove that if $f \sim g$ then $f \asymp g$.

(ii) Prove that if $f \asymp g$, with $\lim_{x\to x_0} \frac{f(x)}{g(x)} = \ell$, then $f \sim \ell g$.

(iii) Give an example showing that $f = \mathcal{O}(g)$ does not imply $f \asymp g$.

(b) Prove that $f \sim g$ if and only if $f = g + o(g)$. *Hint: $f = g + o(g)$ means that $f - g = o(g)$.*

(c) (i) Prove that if $f \sim g$ then $g \sim f$. *Hint: first show, using Theorem 6.2 of the lecture notes, that $f(x) \neq 0$ on a deleted neighborhood of x_0.*

(ii) Prove that if $f \sim g$ and $g \sim h$ then $f \sim h$.

(iii) Prove that if $f = o(g)$ and $g = \mathcal{O}(h)$ then $f = o(h)$.

(d) Let $x_0 \in \mathbb{R}$ and let f be defined on a neighborhood of x_0. Prove that f is continuous at x_0 if and only if $f(x) = f(x_0) + o(1)$ as $x \to x_0$.

(L1) **Series.**

(a) *Geometric series.*

(i) Let $r \in \mathbb{R}$, $r \neq 1$. Show that the partial sums of the geometric series satisfy

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}.$$

(ii) Using a geometric series, write the number $2.3\overline{17} = 2.3171717\cdots$ as a fraction.

(iii) *Bouncing ball.* A ball is dropped from a height of 10 m. Each time it hits the ground, it bounces back up to 60% of the height from which it fell. Find the total distance traveled by the ball. Falling from height h takes $\sqrt{2h/g}$ seconds, and so does rising to height h. Show that the ball stops bouncing after a finite time, and compute this time (take $g = 9.81$ m/s^2).

(iv) *Drug dosing.* A patient takes 100 mg of a drug every 24 hours. After 24 hours, 25% of the amount of drug that was in the body remains. Find the amount of drug in the body just after the n-th dose, and its limit as $n \to \infty$.

(b) *Cauchy Condensation Test.* Let $q > 0$. Use Theorem 7.1 of the lecture notes to show that $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^q}$ converges if and only if $q > 1$.

(c) *The necessary condition.* Show that $\displaystyle\sum_{k=2}^{\infty} (-1)^k \sqrt{\frac{k^3+3}{2k^3-5}}$ does not converge. Why can't Leibniz's Test be applied?

(d) *Comparison and Asymptotic Comparison Tests.* Using one of these tests, decide whether the following series converge:

$$\text{(i)} \sum_{k=1}^{\infty} \frac{1}{2^k - 1} \qquad \text{(ii)} \sum_{k=1}^{\infty} \frac{k+3}{\sqrt[3]{k^9 + k^2}}$$

(e) *Ratio and Root Tests.* Using one of these tests, decide whether the following series converge:

$$\text{(i)} \sum_{n=1}^{\infty} \frac{n!}{n^n} \qquad \text{(ii)} \sum_{n=1}^{\infty} \frac{n^n}{n!} \qquad \text{(iii)} \sum_{n=1}^{\infty} \frac{2^n\, n!}{n^n} \text{ and } \sum_{n=1}^{\infty} \frac{3^n\, n!}{n^n}$$

$$\text{(iv)} \sum_{n=1}^{\infty} \frac{n^3}{3^n} \qquad \text{(v)} \sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$$

(f) *Leibniz's Alternating Series Test.* Show that each of the following series converges. Does it converge absolutely or only conditionally?

$$\text{(i)} \sum_{k=1}^{\infty} \sin\left(k\pi + \frac{1}{k}\right) \qquad \text{(ii)} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k^3+1} \qquad \text{(iii)} \sum_{k=1}^{\infty} (-1)^k \ln\left(1 + \frac{1}{k}\right)$$

Hint for (i): $\sin(k\pi + t) = (-1)^k \sin t$.

(g) *Absolute convergence.*

(i) Does $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 5^{k-1}}{(k+1)^2\, 4^{k+2}}$ converge absolutely? Does it converge?

(ii) Does $\displaystyle\sum_{k=1}^{\infty} \frac{\cos 3k}{k^3}$ converge absolutely?

(L2) **The harmonic series.** For $n \in \mathbb{N}_+$ let $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ be the n-th partial sum of the harmonic series.

(a) *Oresme's proof that the harmonic series diverges.*

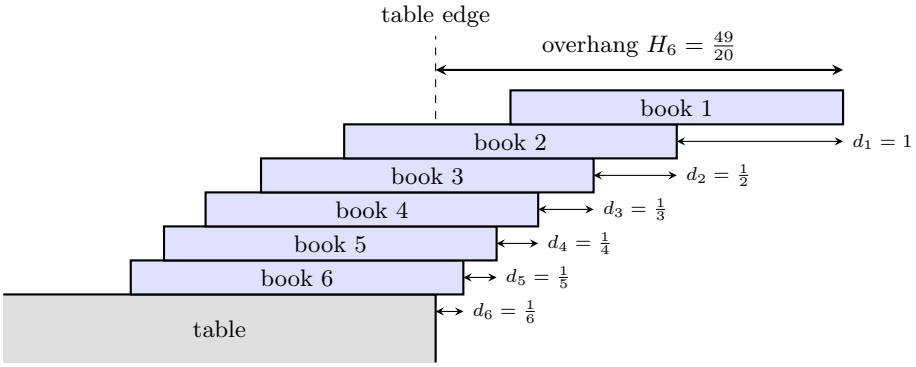
(i) Show that for every $k \in \mathbb{N}_+$, the 2^{k-1} numbers $\frac{1}{2^{k-1}+1}, \frac{1}{2^{k-1}+2}, \ldots, \frac{1}{2^k}$ add up to at least $\frac{1}{2}$.

(ii) Deduce that $H_{2^m} \geq 1 + \frac{m}{2}$ for every $m \in \mathbb{N}$.

(iii) Conclude that $\lim_{n\to\infty} H_n = +\infty$.

(b) The harmonic series diverges very slowly: using (a), find some n for which $H_n \geq 10$.

(c) *Book stacking.* We stack n identical books, each of length 2, at the edge of a table, as in the figure below: book 1 is on top and book n lies on the table. Let d_k be the distance by which the right edge of book k sticks out beyond the right edge of the book below it (for $k = n$: beyond the edge of the table). We use three facts from physics: the center of mass of a single book is at its midpoint; the center of mass of several identical books is the average of their centers of mass; and a stack resting on a support does not tip over as long as its center of mass is not beyond the edge of the support.



(i) Explain why the largest possible value of d_1 is 1, and why in that case the center of mass of book 1 lies exactly above the right edge of book 2.

(ii) Suppose that the center of mass of the top k books lies exactly above the right edge of book $k+1$, and put the origin of the x-axis at this edge. Where is the center of mass of book $k+1$?

(iii) Show that the center of mass of the top $k+1$ books is at $x = -\frac{1}{k+1}$.

(iv) Deduce that the largest possible value of d_{k+1} is $\frac{1}{k+1}$, and that in that case the center of mass of the top $k+1$ books lies exactly above the right edge of book $k+2$ (or of the table, if $k+1 = n$).

(v) Conclude by induction that with $d_k = \frac{1}{k}$ for $k = 1, \ldots, n$ the right edge of the top book is at distance H_n beyond the edge of the table.

(vi) With these values every part of the stack is exactly balanced, so in practice the overhang can only be made arbitrarily close to H_n. Using (a), find a number of books that allows an overhang larger than 3. Then check with a calculator that 11 books suffice, but 10 do not.

(vii) Can the top book be arbitrarily far beyond the edge of the table?